

1.0 – Scenario Background

Paws & Whiskers (P&W's) is a growing pet shop aiming to improve its business through better use of data. Currently, the shop relies on Excel spreadsheets to store sales, customer information, and inventory records, but these will become inefficient and unreliable as the business expands. To support future growth, the company is exploring a transition to Microsoft Azure, allowing data to be stored safely, securely, and consistently. This report outlines how Paws & Whiskers can achieve this by examining relevant data laws, suitable Azure services, required data types, and additional considerations. The next section will look at the data laws they must comply with.

2.0 – Data Laws

2.1 – GDPR (General Data Protection Regulation)

- This law ensures personal data is handled securely and that organisations are accountable for any loss or misuse of personal data (Regulation 2016).
- GDPR is broken into three key areas relevant to Paws & Whiskers:

Lawful Purpose

- Paws & Whiskers must only collect customer data for a clear, lawful reason and should avoid unnecessary information. Examples include names, addresses, pet details, and purchase history (Regulation 2016).

Customer Rights

- Customer data must be accurate, up to date, and stored securely (Regulation 2016). Customers can request access to their data, and they can ask for corrections or deletion. The business must have a simple process for responding to these requests.

Data Minimisation & Security

- Only the minimum amount of data needed should be stored; no unnecessary information should be kept (Regulation 2016). Information must be protected from unauthorised access or misuse. This is achieved through secure passwords, protected devices, and basic staff training to support good data handling practices (Regulation 2016).

2.2 – Data Protection Act (DPA) 2018

This is UK law that works alongside GDPR to govern how personal data is collected, stored, and used in the UK. The DPA affects how Paws & Whiskers must handle customer information in three main areas:

Fair & Lawful Processing

- Customer data must be collected and used fairly and only for legitimate business purposes, with an explanation to the customer about how it will be collected and used (DPA 2018).

Purpose & Storage Limitation

- Data cannot be reused for unrelated purposes without permission (DPA 2018). Customer information must only be kept for as long as necessary and then securely deleted (DPA 2018).

Data Handling Responsibilities

- Staff must follow proper data-handling procedures to prevent misuse or accidental loss (ICO 2024). Any data breach must be reported promptly to the appropriate authority, such as the ICO (ICO 2024).

2.3 – Other Industry Standards (PCI-DSS, PECR, Cybersecurity)

Additional standards apply to Paws & Whiskers because they handle card payments, store customer information, and send marketing communications.

PCI-DSS – Payment Card Industry Data Security Standard (PCI SSC, 2024)

A security standard for any business that processes card payments, ensuring cardholder data is protected.

Requirements for Paws & Whiskers:

- Card data must be encrypted and never stored in plain text.
- POS systems must be secured against unauthorised access.
- Payment information must be handled safely at all times.

PECR – Privacy and Electronic Communications Regulations (ICO, 2026)

A UK regulation controlling how businesses send marketing messages by email, SMS, or phone.

Requirements for Paws & Whiskers:

- Customers must give clear consent before receiving marketing messages.
- The business must keep a record of communication preferences.
- Marketing must stop immediately if a customer withdraws consent.

General Retail Cybersecurity Standards (NCSC, 2026)

Best-practice expectations for protecting customer and business data.

Requirements for Paws & Whiskers:

- Secure POS systems against tampering or misuse.
- Protect loyalty scheme data and customer accounts.
- Avoid collecting unnecessary personal information to reduce risk.

3.0 - Azure Service Recommendations

3.1 – Data Storage

Paws & Whiskers needs a safe, scalable place to store customer details, sales transactions, and inventory data.

3.2 – Azure SQL Database

Stores structured data such as customer records, product lists, and sales transactions (Microsoft, 2024a).

Why Azure SQL Database is the Most Suitable Option

- Centralised and secure database (Microsoft, 2024b).
- Built-in encryption protects customer and sales data (Microsoft, 2024b).
- Automatic backups reduce the risk of data loss (Microsoft, 2024c).
- Scales easily as the shop grows (Microsoft, 2024d).
- Supports relationships (e.g., Customer → Sales), ideal for retail data (Microsoft, 2024a).
- Well-suited for numerical and text-based information, which matches Paws & Whiskers' current data needs (Microsoft, 2024a).
- Handles all structured data the shop uses (customer, sales, inventory) (Microsoft, 2024a).
- Supports GDPR/DPA compliance through encryption and access controls (Microsoft, 2024b).
- Reduces risk by replacing spreadsheets with secure, centralised storage (Microsoft, 2024b).

3.3 – Data Analysis Tools

These tools help the shop understand sales trends and customer behaviour.

3.4 - Azure Synapse Analytics

Analyses sales trends, seasonal demand, and stock performance (Microsoft, 2024f).

Why this is the most suitable option (Microsoft, 2024f):

- Designed for analytics, not day-to-day operations.
- Handles large datasets better than Excel or SQL alone.
- Ideal for dashboards and business reporting.
- Combines data from SQL, Blob Storage, and external sources.

3.5 - Azure Machine Learning

Predicts customer buying patterns and forecasts inventory needs (Microsoft, 2024g).

Why this is the most suitable option (Microsoft, 2024g):

- Provides predictive insights (e.g., which products will sell more).
- Helps reduce over-stocking or under-stocking.
- Supports customer segmentation for targeted marketing.
- Scales with business growth as more data becomes available.

3.6 - Why These Two Meet the Business Requirements

- Synapse provides insights into past performance (sales trends, demand patterns) (Microsoft, 2024f).
- Machine Learning predicts future behaviour (customer habits, stock needs) (Microsoft, 2024g).
- Together they support the business goal of improving reporting, forecasting, and customer understanding (Microsoft, 2024g).
- Both tools analyse data stored securely in Azure, supporting GDPR/DPA expectations.

4.0 Data Integration & Automation

Helps Paws & Whiskers move away from manual spreadsheets.

4.1 - Azure Data Factory

Automates importing data from spreadsheets, POS systems, or supplier files (Microsoft, 2024h).

Why this is the most suitable option:

- Automates data movement — no more manual copying (Microsoft, 2024h).
- Cleans and transforms data before loading it into SQL (Microsoft, 2024h).
- Can run scheduled updates (hourly, daily, weekly) (Microsoft, 2024h).
- Connects to many sources, including online sales platforms (Microsoft, 2024h).

Why Azure Data Factory meets the business requirements:

- Ensures accurate, up-to-date data, supporting GDPR/DPA requirements for data accuracy (Microsoft, 2024h).

- Streamlines operations by reducing manual work and minimising errors (Microsoft, 2024h).
- Maintains reliable, consistent data without extra staff effort (Microsoft, 2024h)

5.0 - Data Types and Data Modelling

Paws & Whiskers works with several key data categories, including customer demographics, transaction history, product and inventory information, pet details, and supplier information. These support the shop's sales, customer management, and stock-tracking needs.

5.1 - Relational Data Model (Operational Use)

A relational model is ideal for day-to-day operations because it keeps data organised, consistent, and easy to query.

5.2 - Core Tables (Entities + Primary Keys)

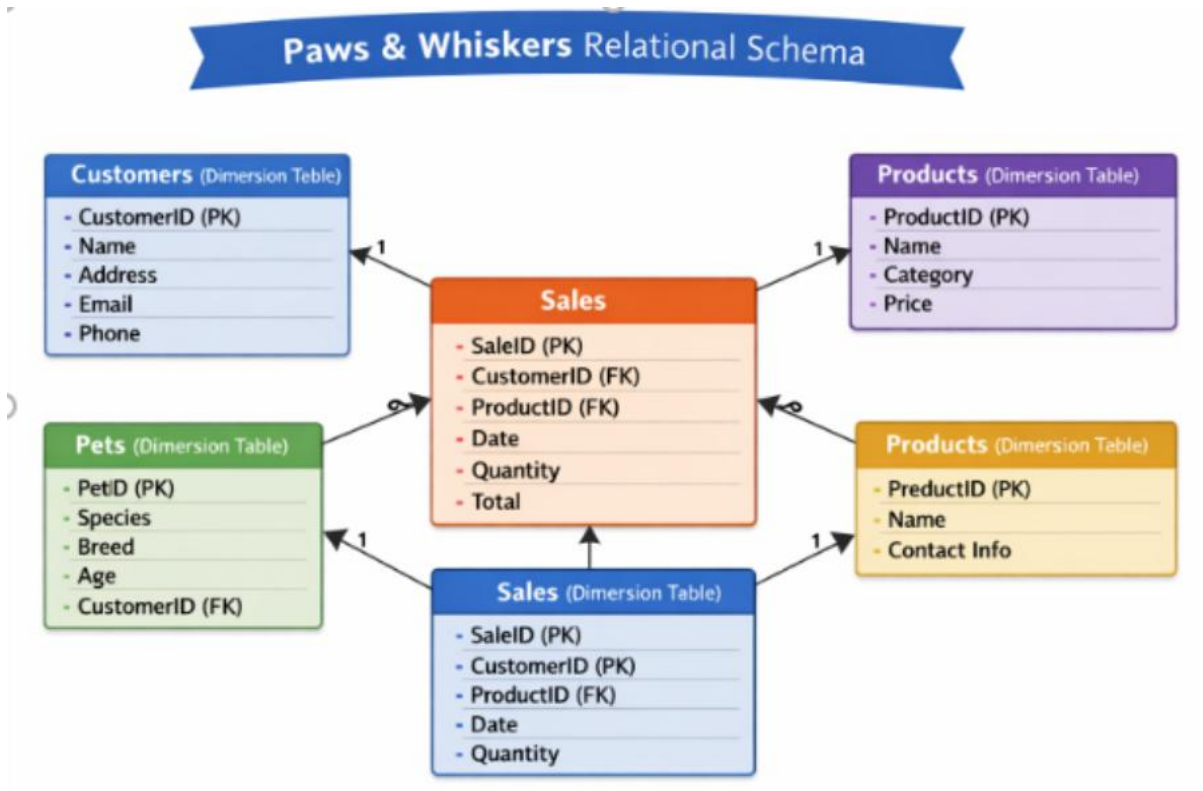
- Customers — CustomerID (PK), Name, Address, Contact details
- Pets — PetID (PK), Species, Breed, Age, CustomerID (FK)
- Products — ProductID (PK), Name, Category, Price, SupplierID (FK)
- Sales — SaleID (PK), CustomerID (FK), ProductID (FK), Date, Quantity, Total
- Suppliers — SupplierID (PK), Name, Contact Info

5.3 - Relationships

- CustomerID → Sales (links customers to their purchases)
- ProductID → Sales (links products to transactions)
- SupplierID → Products (links suppliers to the products they provide)
- CustomerID → Pets (links pets to their owners, if needed)

5.4 - Why This Works for Paws & Whiskers

- Replaces messy spreadsheets with structured, reliable tables.
- Prevents duplicate or inconsistent data.
- Makes it easy to run queries such as:
 - “Top-selling products”
 - “Customers with multiple pets”
 - “Low-stock items”
- Supports accurate reporting and efficient daily operations.



6.0 - Data Storage Formats and Structures in Azure

6.1 - Data Formats

- Paws & Whiskers uses several data formats depending on how information is collected, stored, and analysed.
- Each format supports a different part of the business.

6.2 - CSV – Raw Data Imports

- CSV (Comma-Separated Values) is used for importing simple, table-based data such as customer lists, product inventories, and daily sales.
- Commonly exported from spreadsheets or Point-of-Sale (POS) systems — the tills used in the shop to record purchases.
- Lightweight and easy to edit in Excel.
- Uploads smoothly into Azure SQL Database (Microsoft, 2024i).
- Ideal for preparing operational data before it enters the cloud.

6.3 - JSON – Structured Data

- JSON (JavaScript Object Notation) is used when Paws & Whiskers needs to transfer structured data between systems.

- Examples include customer profiles from the website or online order details.
- Works well with Azure Data Factory for automated data movement (Microsoft, 2024j).
- Human-readable structure makes it suitable for supplier integrations and loyalty-scheme platforms.

6.4 - Parquet – Analytical Data

- Parquet is a column-based format designed for analytics.
- Used to store large datasets such as long-term sales history, seasonal demand patterns, and customer purchasing trends.
- Highly compressed and optimised for Azure Synapse Analytics (Microsoft, 2024k) and Power BI.
- Enables fast queries and efficient reporting.

6.5 - Data Security and Encryption (with in-text citations)

- Paws & Whiskers must protect customer and transaction data in line with GDPR and the Data Protection Act (DPA) 2018.
- Azure includes built-in security features that help the business keep information safe and compliant.

6.6 - Encryption at Rest and in Transit

- All data stored in Azure is automatically encrypted using AES-256 (Microsoft, 2024l).
- Data moving between systems is protected using HTTPS and Transport Layer Security (TLS) (Microsoft, 2024m).
- Ensures customer details, payment information, and sales records remain secure whether stored or transferred.

6.7 - Access Controls (RBAC)

- Role-Based Access Control (RBAC) ensures only authorised staff can access specific data (Microsoft, 2024n).
- Managers can view sales reports, while shop assistants can update stock but cannot access customer contact details.
- Supports GDPR principles such as data minimisation.
- Prevents unauthorised access.

6.8 - Azure Key Vault

- Azure Key Vault securely stores passwords, encryption keys, and API tokens (Microsoft, 2024o).

- Prevents credentials from being exposed in code, spreadsheets, or emails.
- Supports GDPR and PCI-DSS compliance by ensuring sensitive information is stored safely and accessed only by authorised systems.

7.0 - Additional Considerations

7.1 - Backup & Disaster Recovery

7.2 - Azure Backup

- Protects customer, sales, and inventory data from accidental deletion or corruption (Microsoft, 2024p).
- Creates restore points so Paws & Whiskers can quickly recover lost data (Microsoft, 2024p).
- Supports business continuity by ensuring daily operations can continue even after data issues.

7.3 - Azure Site Recovery

- Replicates key systems to another Azure region (Microsoft, 2024q).
- Keeps sales and stock systems running during outages or hardware failures (Microsoft, 2024q).
- Helps the business meet GDPR/DPA requirements for data availability and resilience.

7.4 - Data Visualisation

7.5 - Power BI

- Connects directly to Azure SQL Database and Azure Synapse Analytics (Microsoft, 2024r).

- Produces real-time dashboards showing sales performance, low-stock alerts, and customer trends as seen below (Microsoft, 2024r).



- Helps managers make fast, data-driven decisions — such as identifying best-selling products or spotting seasonal demand.

7.6 - Future Scalability

7.7 - Azure Scalability Features

- Automatically scales storage and computing power as the business grows (Microsoft, 2024s).
- Handles larger datasets if Paws & Whiskers expands to multiple stores or adds new services (Microsoft, 2024s).
- Pay-as-you-go pricing keeps costs manageable while supporting long-term growth.

8.0 - Conclusion

Paws & Whiskers can greatly improve how it manages data by moving from spreadsheets to Microsoft Azure. The business will gain secure, compliant storage, automated data processing, and stronger analytical tools for better decision-making. By following GDPR and DPA requirements and using structured Azure services, the shop can keep its data accurate, protected, and ready for growth. Azure's scalability, backup options, and visualisation tools also provide a solid foundation for long-term development as the business expands.

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